SUB-MODULE 7.4 · REINFORCEMENT LEARNING FROM HUMAN FEEDBACK

RLHF Overview

The 3-stage pipeline (SFT → RM → PPO) that aligned GPT-4, Claude, Gemini, and Llama 3

9 min	B2 – Understand	L3 – Advanced	AI-RL-IN-TH-000001
Duration	Bloom's	Level	ID

RLHF — Definition and Historical Context

RLHF: Reinforcement Learning from Human Feedback

A three-stage alignment pipeline that trains language models to produce outputs that humans prefer. Stage 1: SFT baseline. Stage 2: Reward model from human preferences. Stage 3: RL optimisation of the policy to maximise reward.

- Origin: InstructGPT (Ouyang et al., NeurIPS 2022) — the paper that established RLHF as the standard for LLM alignment
- Impact: GPT-3 → InstructGPT showed that a 1.3B RLHF model was preferred over the raw 175B GPT-3 by human evaluators
- Adopted by: GPT-4 (OpenAI), Claude (Anthropic), Gemini (Google), Llama 3 (Meta), Qwen (Alibaba)
- Why it matters: pre-training optimises for next-token prediction — not helpfulness. RLHF bridges this gap.

The 3-Stage RLHF Pipeline

1

Stage 1
SFT Baseline

Fine-tune base model on high-quality human demonstrations. Creates a well-behaved starting policy π_SFT.

2

Stage 2
Reward Model

Collect pairwise human preferences (which response is better?). Train RM to predict human preference scores r(x,y).

3

Stage 3
PPO Optimisation

Use PPO to optimise π_θ to maximise RM reward, subject to KL constraint vs π_SFT. Prevents reward hacking.

Why RLHF Changed Everything — GPT-3 vs InstructGPT

GPT-3 (Pre-trained only)

- Optimised for: next-token prediction
- Prompt: 'List ways to improve my essay'
- Responds with: more essay improvement prompts
- Doesn't understand the user's intent
- Needs heavy prompt engineering
- Often ignores instruction framing

InstructGPT (SFT + RLHF)

- Optimised for: human preference
- Prompt: 'List ways to improve my essay'
- Responds with: 5 concrete improvement tips
- Understands and follows the instruction
- Works with natural language instructions
- Preferred 71% of the time over GPT-3 175B

RLHF — The Four Model Copies Required

Model	Role	Frozen?	Memory
Actor (π_θ)	The policy being optimised — generates responses	No — updated by PPO	Full model VRAM
Critic	Value function — estimates expected future reward	No — updated alongside actor	Full model VRAM
Reference (π_{SFT})	KL baseline — provides $\pi_{SFT}(a\|s)$ for KL divergence	Yes — frozen	Full model VRAM
Reward Model	Scores responses — provides $r(x,y)$ signal	Yes — frozen	Smaller (up to full model)

KEY REFERENCES

- Ouyang, L. et al. (2022). Training LMs to Follow Instructions with Human Feedback (InstructGPT). NeurIPS 2022.
- Lambert, N. (2024). RLHF Book. rlhfbook.com
- Ziegler, D.M. et al. (2019). Fine-Tuning Language Models from Human Preferences. arXiv.

▶️ **Next: Concept 20: Reward Model Training (AI-RL-IN-TH-000002)**